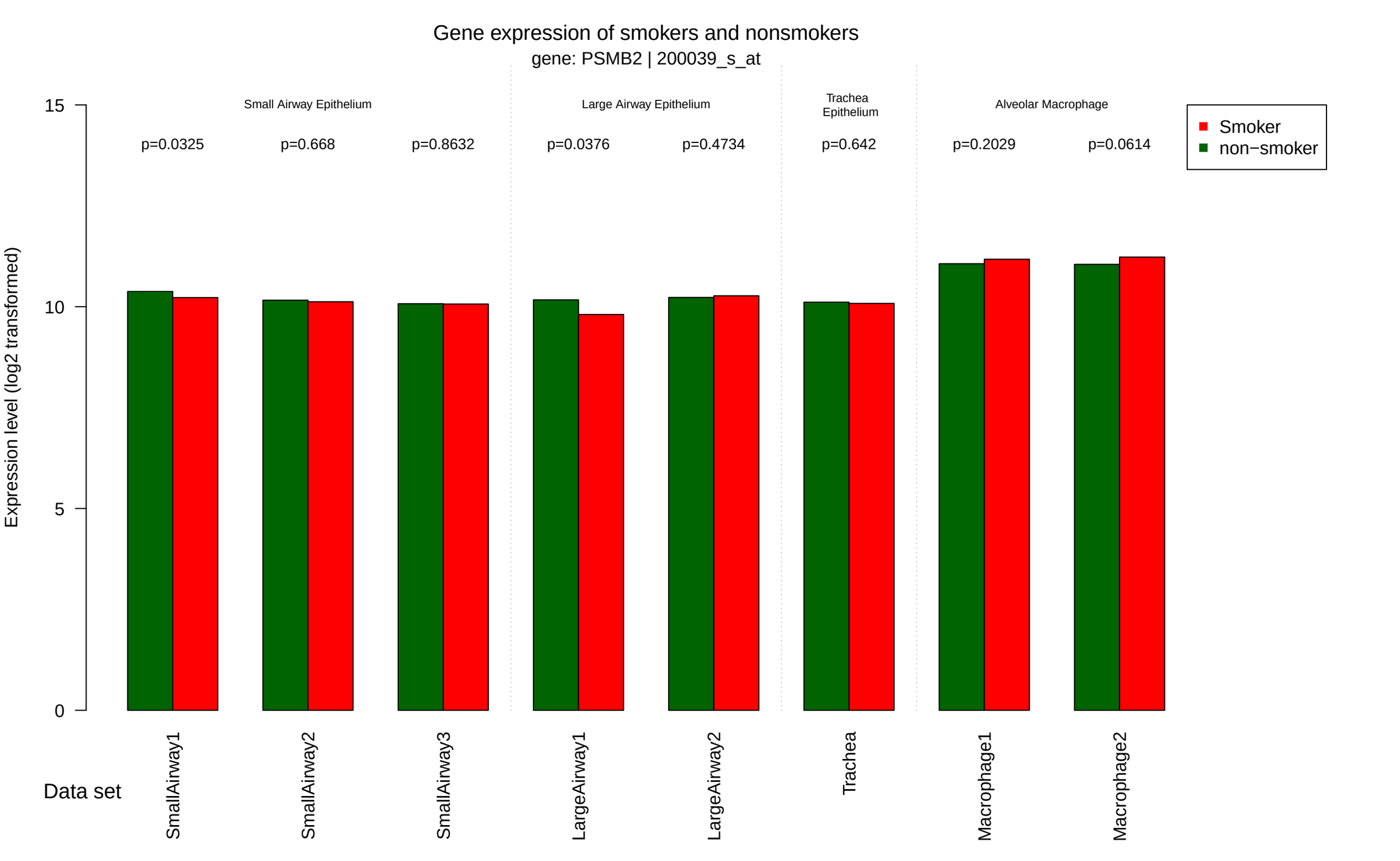
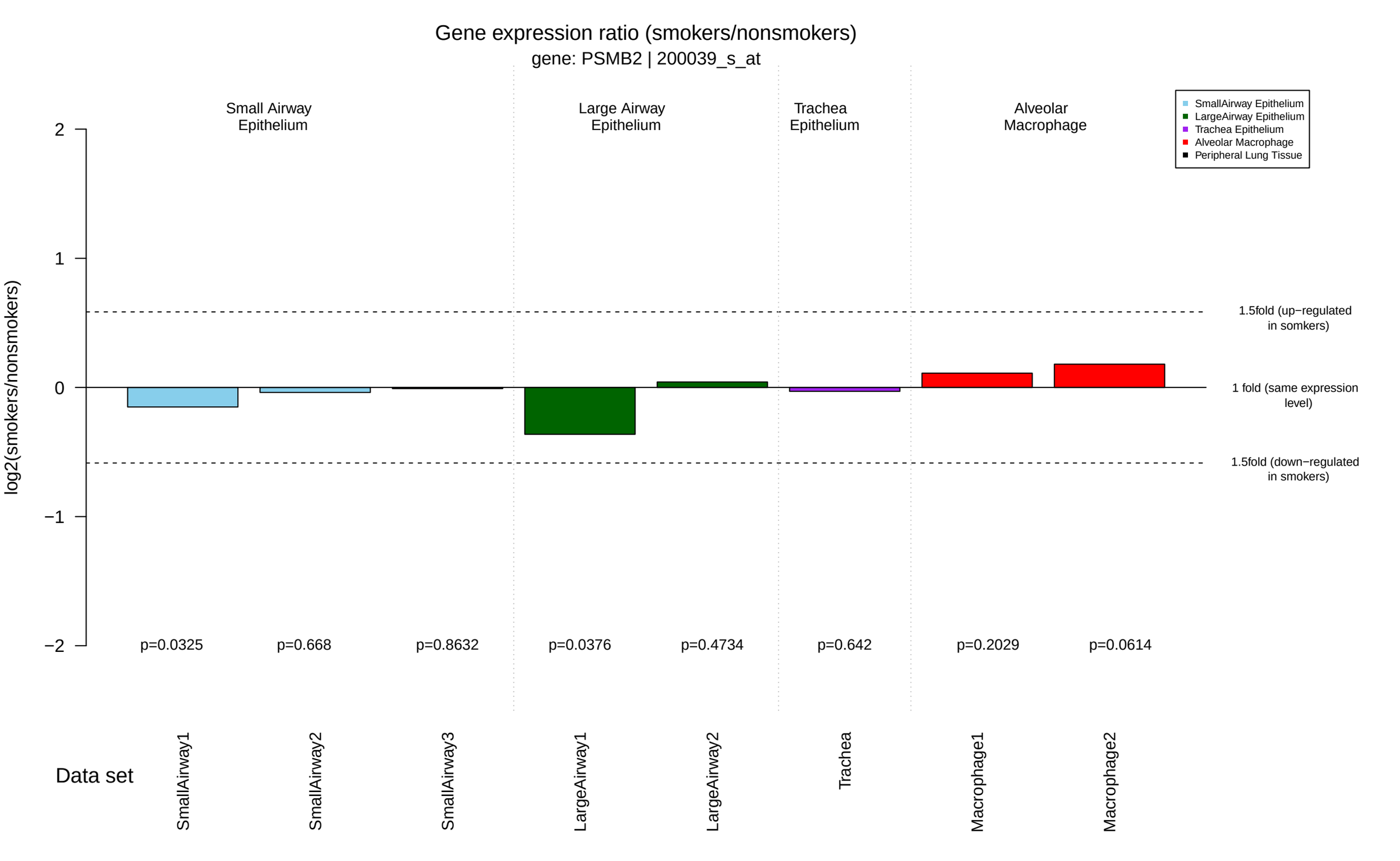
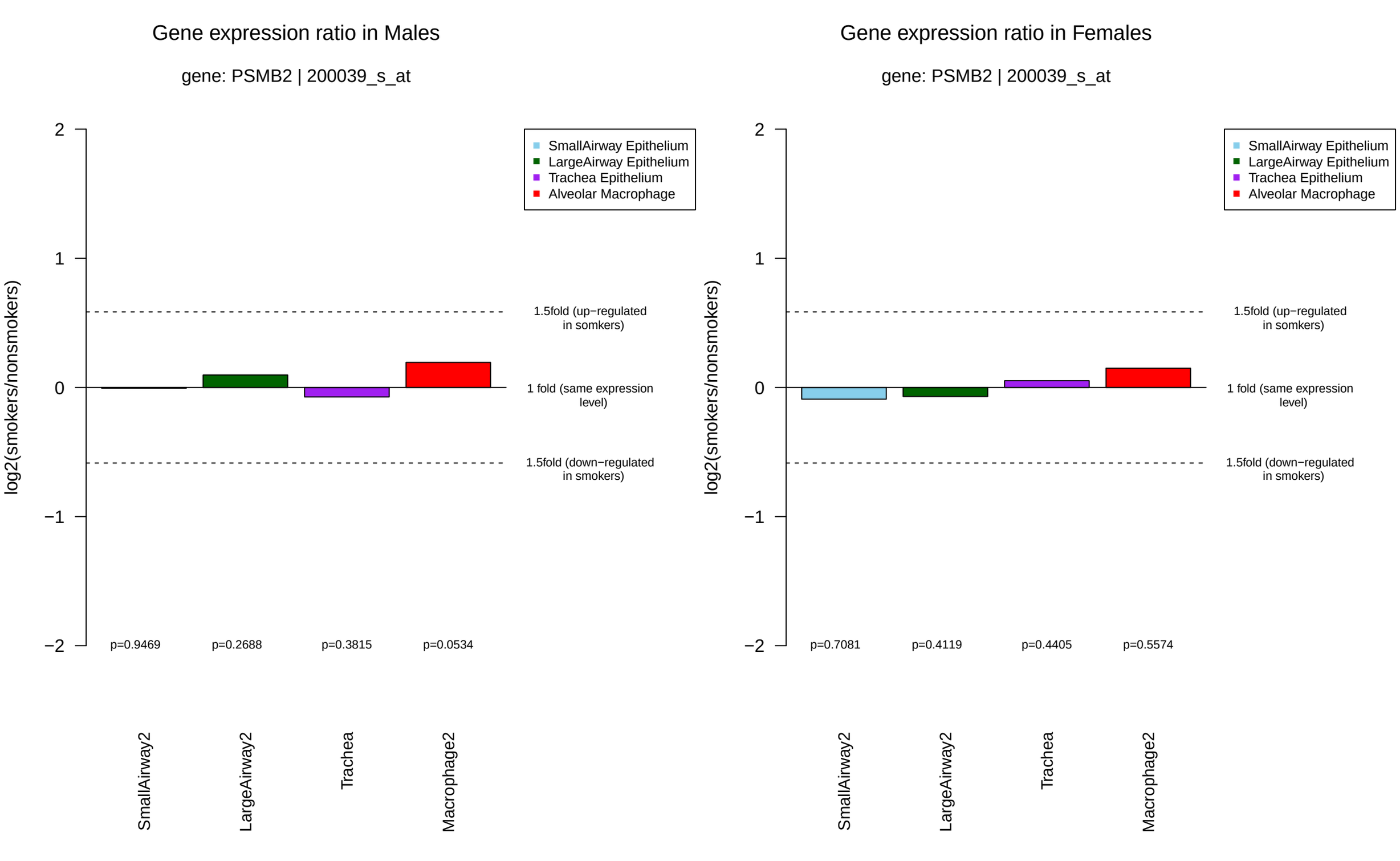
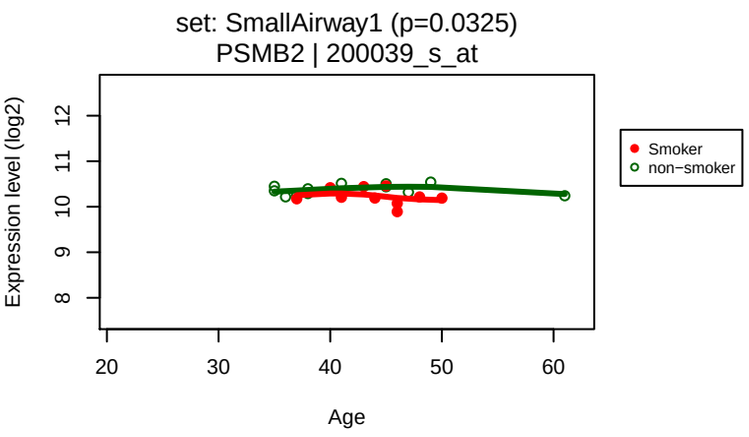
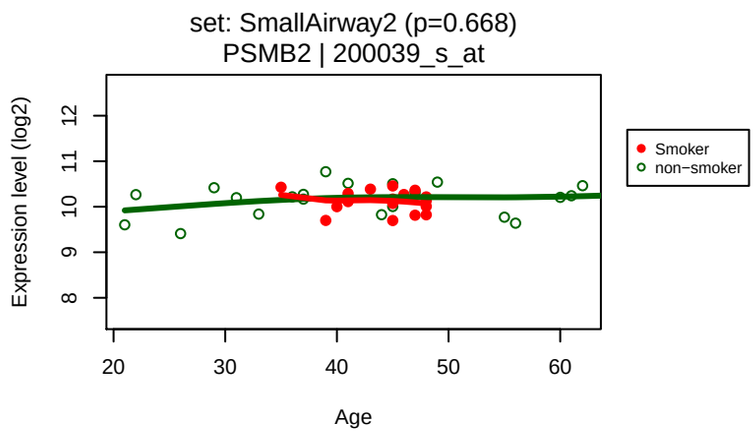
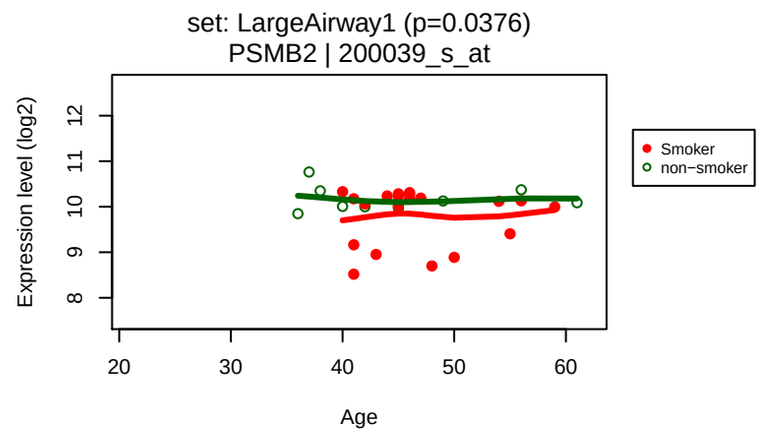

Gene expression of smokers and nonsmokers
gene: PSMB2 | 200039_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0325
p=0.668
p=0.8632
p=0.0376
p=0.4734
p=0.642
p=0.2029
p=0.0614
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

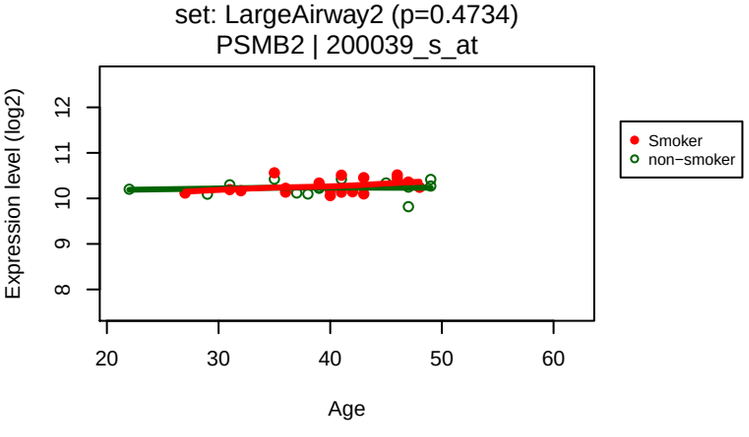

Gene expression ratio (smokers/nonsmokers)
gene: PSMB2 | 200039_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0325
p=0.668
p=0.8632
p=0.0376
p=0.4734
p=0.642
p=0.2029
p=0.0614
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

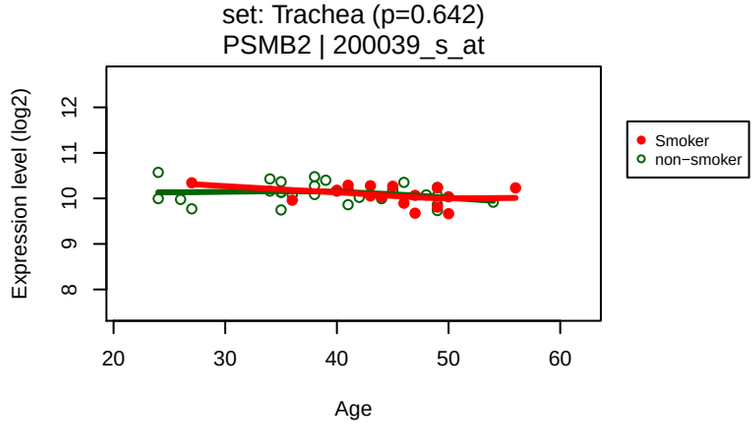

Gene expression ratio in Males
gene: PSMB2 | 200039_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9469
p=0.2688
p=0.3815
p=0.0534
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSMB2 | 200039_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7081
p=0.4119
p=0.4405
p=0.5574
SmallAirway2
LargeAirway2
Trachea
Macrophage2

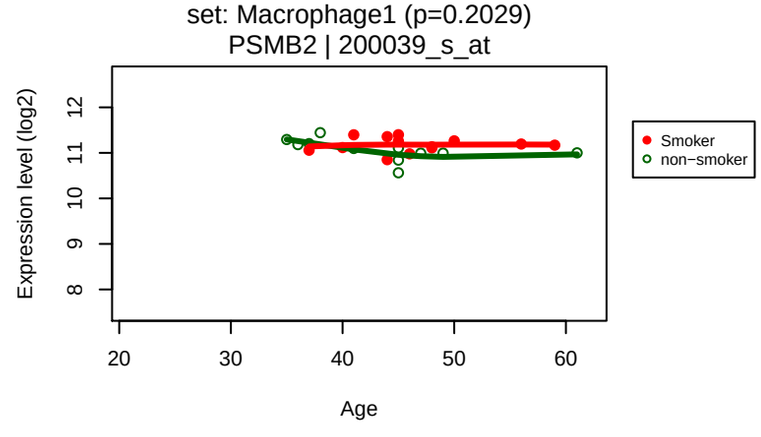

set: SmallAirway1 (p=0.0325)
PSMB2 | 200039_s_at
Expression level (log2)
Age
Smoker
non−smoker

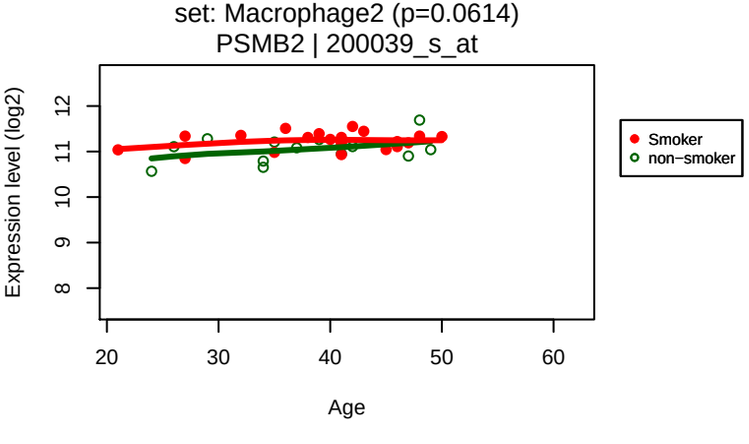

set: SmallAirway2 (p=0.668)
PSMB2 | 200039_s_at
Expression level (log2)
Age
Smoker
non−smoker

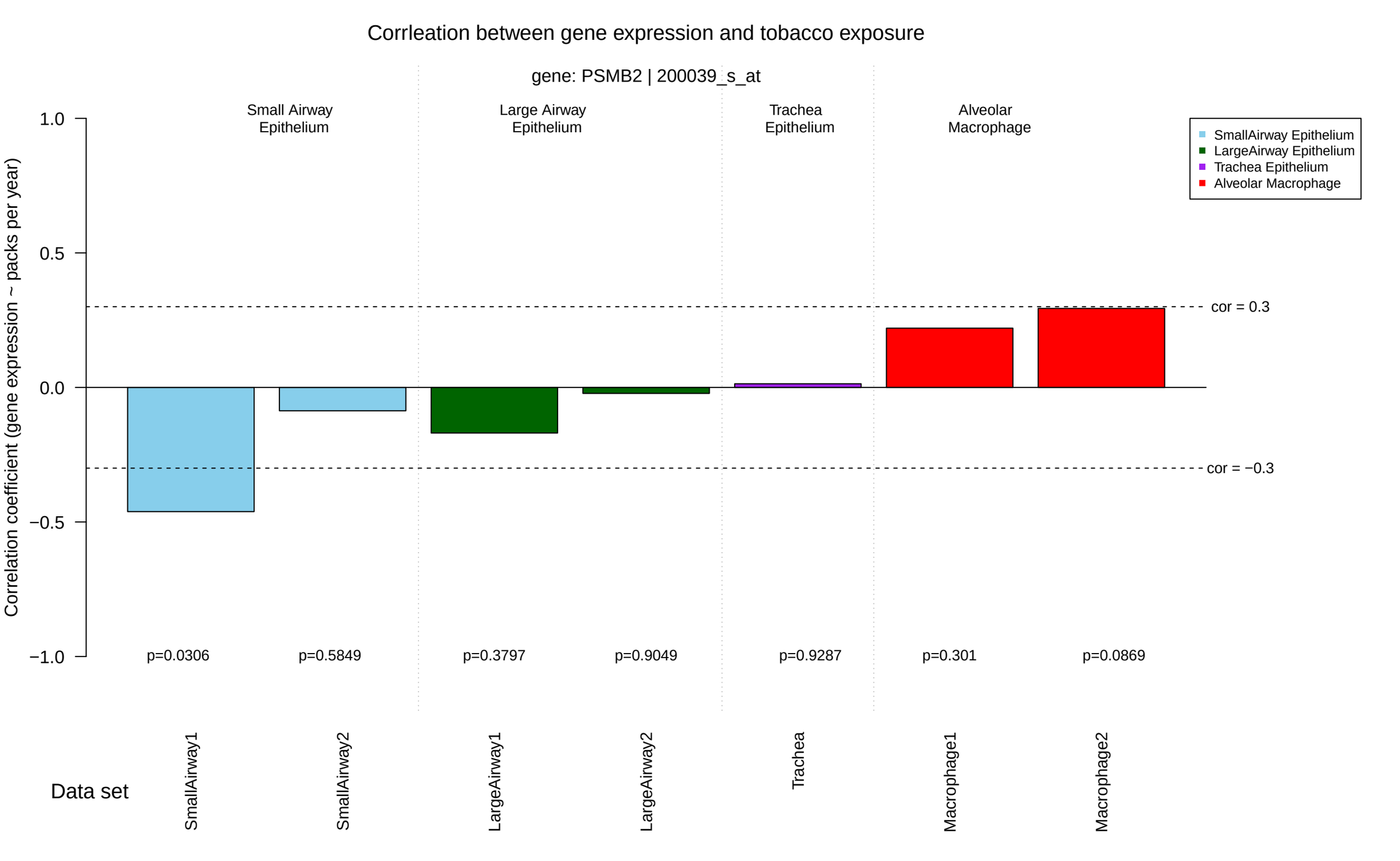

set: LargeAirway1 (p=0.0376)
PSMB2 | 200039_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4734)
PSMB2 | 200039_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.642)
PSMB2 | 200039_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2029)
PSMB2 | 200039_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0614)
PSMB2 | 200039_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSMB2 | 200039_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0306
p=0.5849
p=0.3797
p=0.9049
p=0.9287
p=0.301
p=0.0869
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2